

Financial analysis

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1. Introduction

This report presents an analysis of a financial risk dataset to understand loan performance, identify default patterns, and evaluate key risk indicators. The objective is to uncover insights that can support data-driven decision-making in financial institutions.

2. Load Libraries

```
library(tidyverse)
library(ggplot2)
library(dplyr)
```

3. Load Dataset

```
data <- read.csv("Credit_Risk_Dataset.csv")

head(data)
str(data)
summary(data)
```

4. Data Cleaning

```
# Rename columns for clarity
data <- data %>%
  rename(
    Age = person_age,
    Income = person_income,
    Home_Ownership = person_home_ownership,
    Employment_Length = person_emp_length,
    Loan_Intent = loan_intent,
    Loan_Grade = loan_grade,
    Loan_Amount = loan_amnt,
    Interest_Rate = loan_int_rate,
    Loan_Status = loan_status,
    Loan_Percent_Income = loan_percent_income,
```

```

    Default_History = cb_person_default_on_file,
    Credit_History_Length = cb_person_cred_hist_length
)

```

Convert Loan Status to factor

```

data$Loan_Status <- ifelse(data$Loan_Status == 1, "Default", "Paid")
data$Loan_Status <- as.factor(data$Loan_Status)

```

5. Exploratory Data Analysis

5.1 Loan Status Distribution

```

ggplot(data, aes(x = Loan_Status, fill = Loan_Status)) +
  geom_bar() +
  labs(title = "Loan Status Distribution", x = "Status", y = "Count") +
  theme_minimal()

```

5.2 Loan Amount by Purpose

```

ggplot(data, aes(x = Loan_Purpose, y = Loan_Amount, fill = Loan_Purpose)) +
  geom_bar(stat = "summary", fun = "sum") +
  labs(title = "Total Loan Amount by Purpose", x = "Loan Purpose", y = "Total Loan Amount") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```

5.3 Loan Amount by Risk Grade

```

ggplot(data, aes(x = Loan_Grade, y = Loan_Amount, fill = Loan_Grade)) +
  geom_bar(stat = "summary", fun = "sum") +
  labs(title = "Loan Amount by Risk Grade", x = "Grade", y = "Total Loan Amount") +
  theme_minimal()

```

5.4 Income vs Loan Amount (Risk Analysis)

```

ggplot(data, aes(x = Income, y = Loan_Amount, color = Loan_Status)) +
  geom_point(alpha = 0.6) +
  labs(title = "Income vs Loan Amount", x = "Income", y = "Loan Amount") +
  theme_minimal()

```

5.5 Default Rate by Home Ownership

```

default_rate <- data %>%
  group_by(Home_Ownership) %>%
  summarise(Default_Rate = mean(Loan_Status == "Default"))

ggplot(default_rate, aes(x = Home_Ownership, y = Default_Rate, fill = Home_Ownership)) +
  geom_bar(stat = "identity") +
  labs(title = "Default Rate by Home Ownership", x = "Home Ownership", y = "Default Rate") +
  theme_minimal()

```

6. Key Insights

- Loan defaults represent a significant portion of total loans.
- Certain loan purposes contribute more heavily to total loan amounts.
- Higher loan grades (riskier categories) are associated with larger loan volumes.
- Customers with lower income tend to take proportionally higher loans, increasing risk.
- Default rates vary across home ownership categories, indicating differences in financial stability.

7. Conclusion

This analysis highlights important financial risk patterns within the dataset. By leveraging data visualization and statistical summaries, organizations can better understand borrower behavior and identify high-risk segments. These insights can support improved lending decisions, risk management strategies, and financial planning.